1. J_i, K_i $\quad N_i = \frac{J_i - iK_i}{2}$ $\quad N_i^\dagger = \frac{J_i + iK_i}{2}$

$[N_i, N_j] = i\varepsilon_{ijk} N_k$, $\quad [N_i^\dagger, N_j^\dagger] = i\varepsilon_{ijk} N_k^\dagger$, $\quad [N_i, N_j^\dagger] = 0$.

$n = 2j+1$ $\qquad$ $n' = 2j'+1$

$(2j+1, 2j'+1)$

$(1,1)$: 标量. $\varphi(x)$

$(2,1)$: 左手旋量. $\psi_a(x)$ $\quad a = 1, 2$.

$(1,2)$: 右手旋量. $\chi_{\dot a}^\dagger(x)$ $\quad \dot a = 1, 2$.

$$(\psi_a)^\dagger = \psi_{\dot a}^\dagger$$

$(2,2)$: 矢量表示 $\quad A_{a\dot a}$, $\quad \sigma^\mu_{a\dot a} = (1, \sigma^i)$.

$$\downarrow$$

$$A^\mu = A_{a\dot a}\, \sigma^\mu_{b\dot b}\, \varepsilon^{ab}\, \varepsilon^{\dot a \dot b}$$

2. $(2,1)$ $\psi_{\dot a}$ $\quad U^{-1}\psi_{\dot a}U = \Lambda_{\dot a}{}^{\dot b}(\Lambda)\psi_{\dot b}(\Lambda^{-1}x)$.

$$L^{\mu\nu} \propto x^\mu \partial^\nu - x^\nu \partial^\mu$$

$$\Lambda_a{}^b = \delta_a{}^b + \frac{i}{2}\delta\omega_{\mu\nu}(S_L^{\mu\nu})_a{}^b$$

$$S_L^{ij} = -\frac{1}{2}\epsilon^{ijk}\sigma^k$$

$$S_L^{i0} = -i\frac{1}{2}\sigma^i$$

$$R_{\dot a}{}^{\dot b} = S_{\dot a}{}^{\dot b} + \frac{1}{2}\delta\omega_{\mu\nu}(S_R^{\mu\nu})_{\dot a}{}^{\dot b}$$

$$(S_R^{\mu\nu})_{\dot a}{}^{\dot b} = -[(S_L^{\mu\nu})_a{}^b]^*$$

3. ε^{ab}, $\varepsilon^{\dot{a}\dot{b}}$ $\varepsilon^{ab}\psi_a\chi_b$, $\varepsilon^{\dot{a}\dot{b}}\psi_{\dot{a}}\chi_{\dot{b}}$

Def $\psi\chi \equiv \psi^a\chi_a = \varepsilon^{ab}\psi_b\chi_a$

$$\psi_a\chi^a = -\psi^a\chi_a = -\varepsilon^{ba}\psi_b\chi_a = \psi_b\chi^b$$

这与度规不同，度规为对称的，ε^{ab} 为反对称.

$$\psi^\dagger\chi^\dagger \equiv \psi^\dagger_{\dot{a}}\chi^{\dagger\dot{a}} = -\psi^{\dagger\dot{a}}\chi^\dagger_{\dot{a}}$$

$$(\psi\chi)^\dagger = \chi^\dagger\psi^\dagger$$

Grassman 数： $\psi_a\chi_b = -\chi_b\psi_a$

$$\psi\chi = \chi\psi$$

4. $\sigma^{\mu}_{a\dot{a}}$ $\quad A_{a\dot{a}} \longrightarrow A^{\mu}$

Fierz id : $\quad g_{\mu\nu}\,\sigma^{\mu}_{a\dot{a}}\,\sigma^{\nu}_{b\dot{b}} = 2\,\varepsilon_{ab}\,\varepsilon_{\dot{a}\dot{b}}$

5. $(S_L^{\mu\nu})_a{}^b = -\frac{i}{4}(\sigma^{\mu}\bar{\sigma}^{\nu} - \sigma^{\nu}\bar{\sigma}^{\mu})_a{}^b$

$(S_R^{\mu\nu})^{\dot{a}}{}_{\dot{b}} = \frac{i}{4}(\bar{\sigma}^{\mu}\sigma^{\nu} - \bar{\sigma}^{\nu}\sigma^{\mu})^{\dot{a}}{}_{\dot{b}}$

$\bar{\sigma}^{\mu\dot{a}b} = \varepsilon^{bc}\,\varepsilon^{\dot{a}\dot{b}}\,\sigma^{\mu}{}_{c\dot{b}} = (1, -\sigma^{i})$.

6. $\mathcal{L} = i\psi^{\dagger}\bar{\sigma}^{\mu}\partial_{\mu}\psi - \frac{1}{2}m(\psi\psi + \psi^{\dagger}\psi^{\dagger})$.

补上指标: $i\psi^{\dagger}\bar{\sigma}^{\mu\dot{a}b}\partial_{\mu}\psi_b - \frac{1}{2}m(\psi^{a}\psi_{a} + \psi^{\dagger}_{\dot{a}}\psi^{\dagger\dot{a}})$